Basebot velocity control – PI controller design

DTU Digital Control · Robot Kim (74), left motor · continuous-time design, verified in Simulink

1. Plant

Measured motor parameters (wheels off the floor). J_{total} includes the robot mass (0.95 kg) reflected to the motor shaft. The 10 ms velocity measurement delay is modelled with a 2nd-order Padé approximation:

R	L	$K_b = K_t$	J_{total}	D	Delay T
4.21 Ω	5.6 mH	0.0081 V·s/rad	$5.69 \cdot 10^{-6}$ kg·m^2	$1.06 \cdot 10^{-6}$ N·m·s	10 ms

$G_{tot}(s) = K_t / ((Ls+R)(Js+D) + K_tK_b) \cdot \text{Padé}_2(s)$. The phase reaches −180° at **ω_{180} = 141 rad/s** with $|G_{tot}| = 2.37$, so a P-controller is only stable for $K_p < 0.42$.

2. Controller

Structure: PI, $C(s) = K_p(\tau_i s + 1)/(\tau_i s)$. The plant is type 0, so a P-controller leaves a step error of $1/(1 + 115K_p)$ plus the friction and voltage offsets; the integrator removes it. No D/lead term: the encoder velocity is coarsely quantised and a derivative would amplify that noise into the motor voltage, while the 10 ms delay limits the bandwidth anyway.

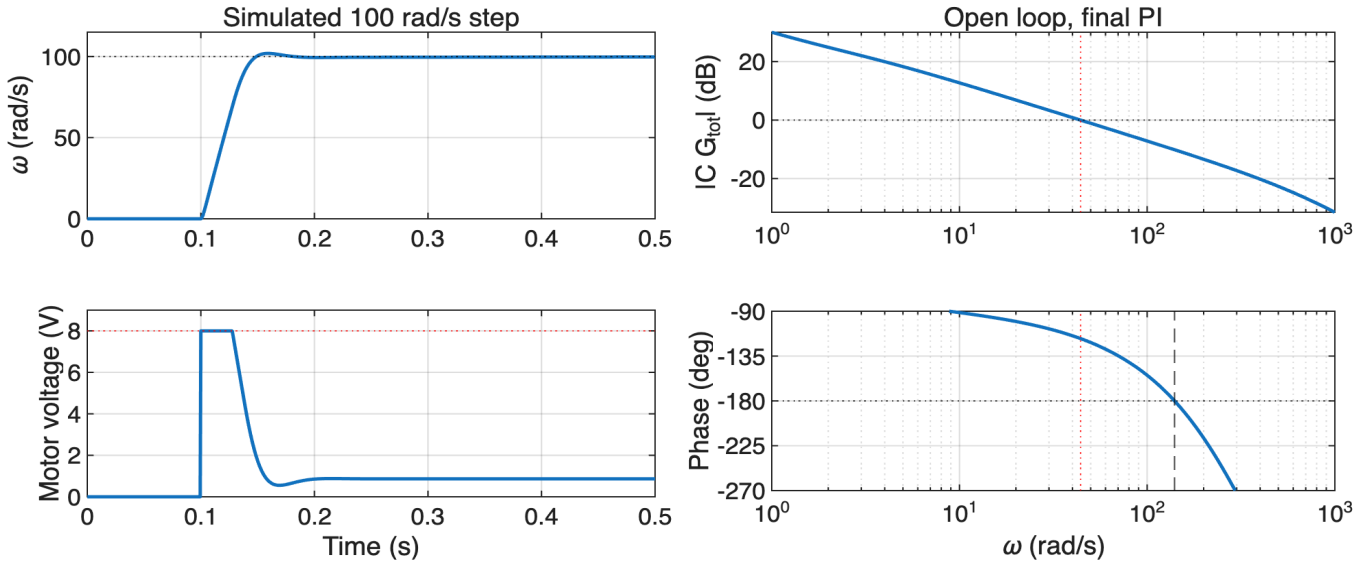
Frequency-domain design: 60° phase margin with the PI zero $N_i = 3$ below crossover gives $\omega_c = 27$ rad/s, $\tau_i = 0.11$ s, $K_p = 0.077$. In simulation this overshoots 14 % and settles in 290 ms: with the robot mass the plant is almost an integrator, so a PI zero close to ω_c adds overshoot, and the ±8 V saturation makes it worse.

Tuned in Simulink (K_p/τ_i sweep; model with 10 ms transport delay on the measured velocity, ±8 V saturation on voltage and integrator, 100 rad/s step at 0.1 s): a higher gain and a slower integrator (zero at 2 rad/s) give the best step response.

Final controller: K_p = 0.13 V/(rad/s), τ_i = 0.5 s

3. Performance (simulated)

Crossover ω_c	Phase margin	Gain margin	Overshoot	Rise (10–90 %)	Settling 2 %	Steady-state error
44 rad/s	63°	10.2 dB	1.9 %	33 ms	45 ms	0



Left: 100 rad/s step, motor velocity and voltage. The voltage is at the 8 V limit for the first 28 ms; it then settles at 0.87 V. Right: open loop $C \cdot G_{tot}$, crossover (red) and −180° frequency (dashed).